Computer Science & IT

Theory of Computation

Regular Languages

Lecture No. 4

By- Ravindra Sir

Recap of Previous Lecture

Topic

Reg languages

Topic

Topic

Topics to be Covered

Topic

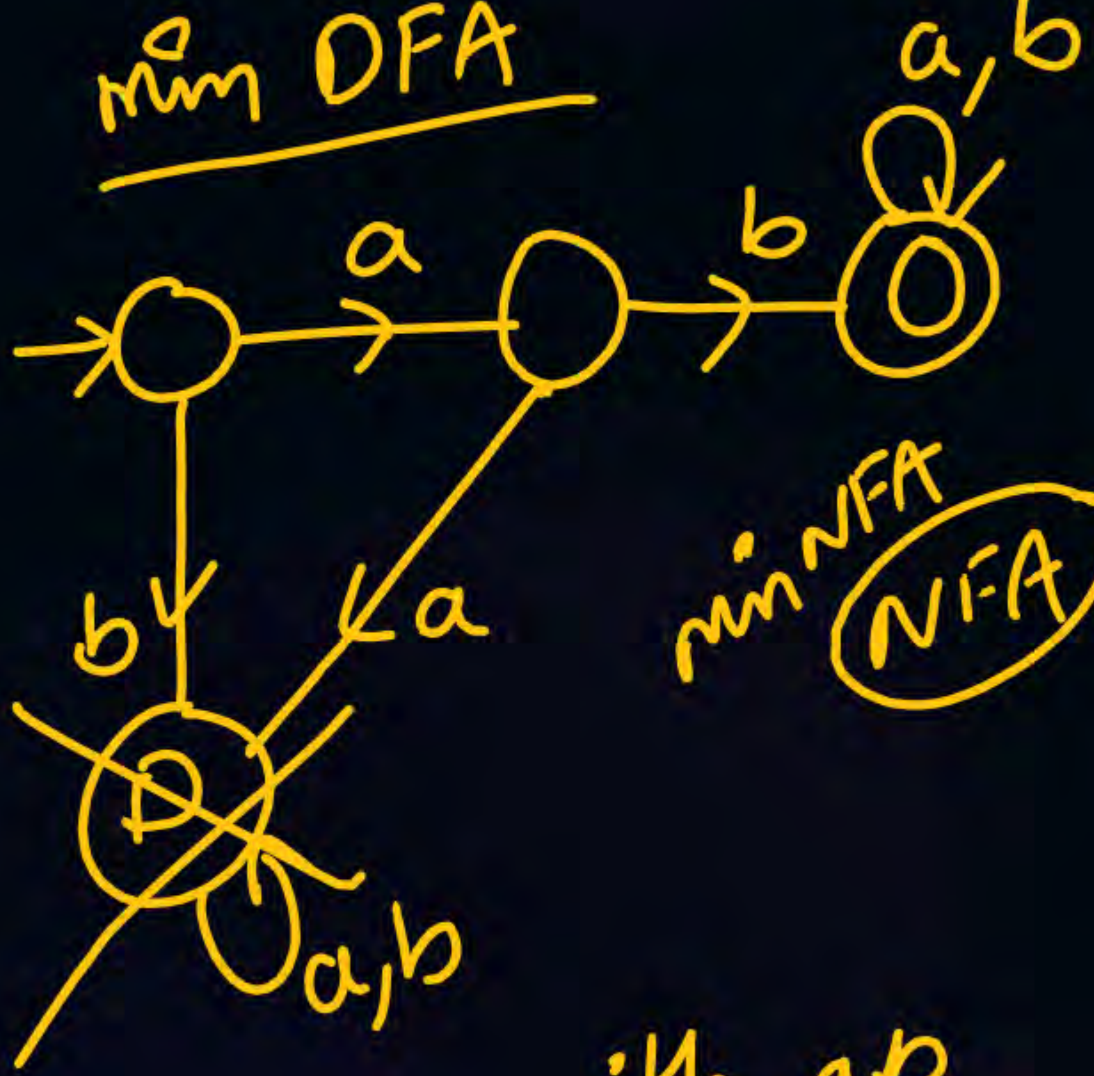
Regular

Topic

Topic

DFA or NFA $\longrightarrow$ language?

min DFA



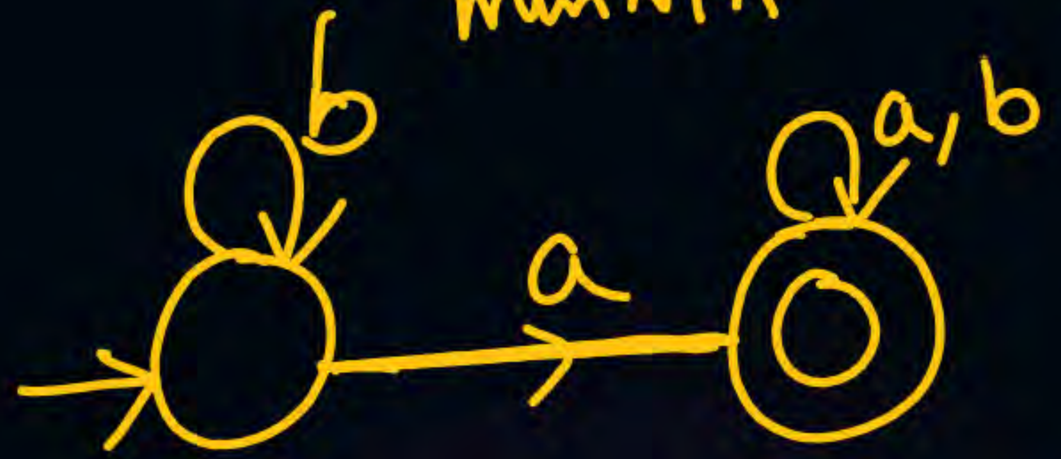
start with ab

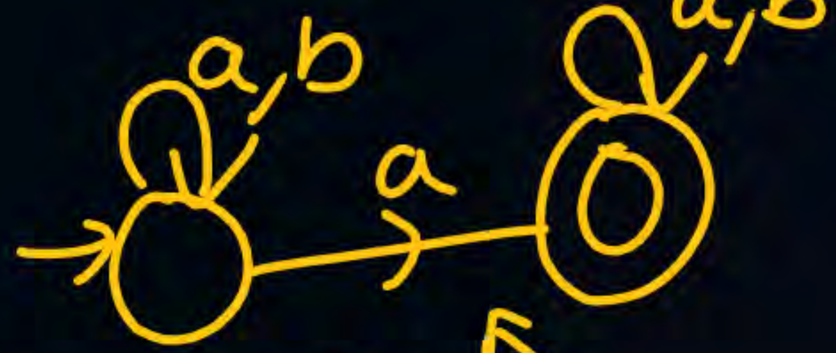
min NFA

NFA → ?

$ab(a+b)^*$

$\frac{\text{min DFA}}{\text{min NFA}}$





min NFA

min NFA

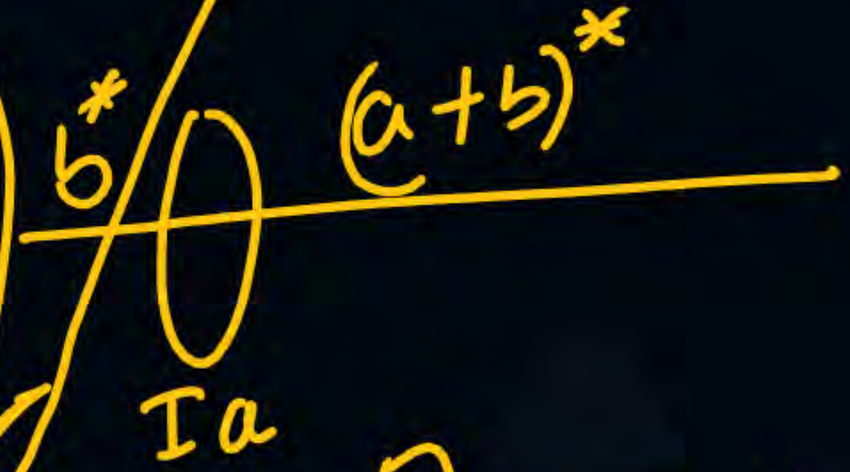
✓ atleast one 'a'

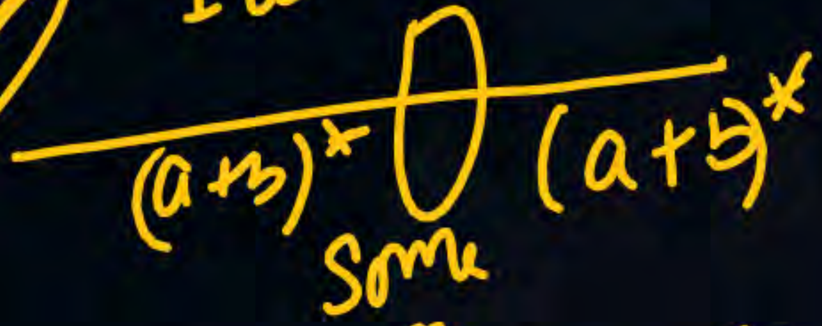
✓ Containing a 'a'

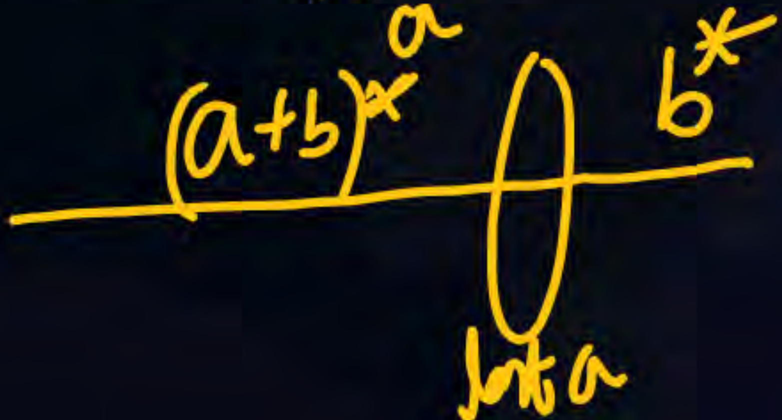
$b^* a (a+b)^*$

$(a+b)^* a (a+b)^*$ ✓

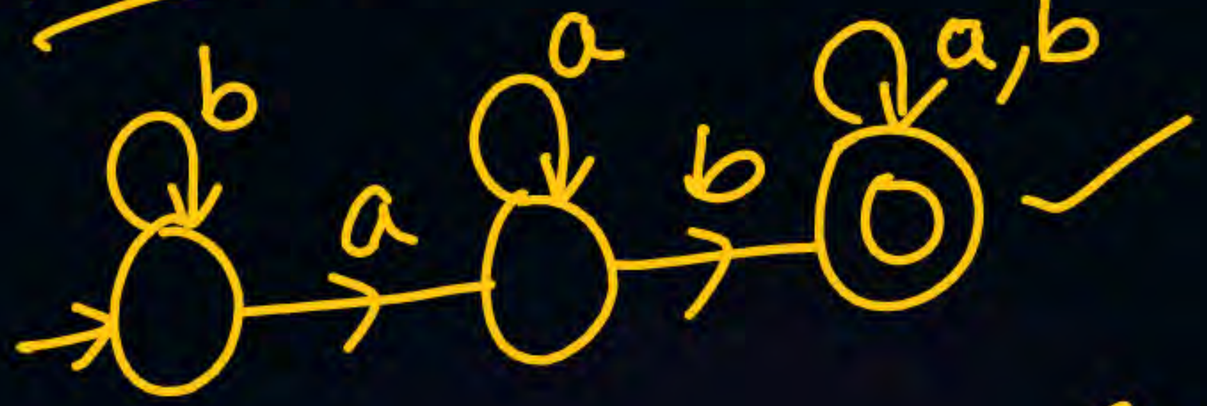
$(a+b)^* a b^*$







min DFA :-   min NFA :-

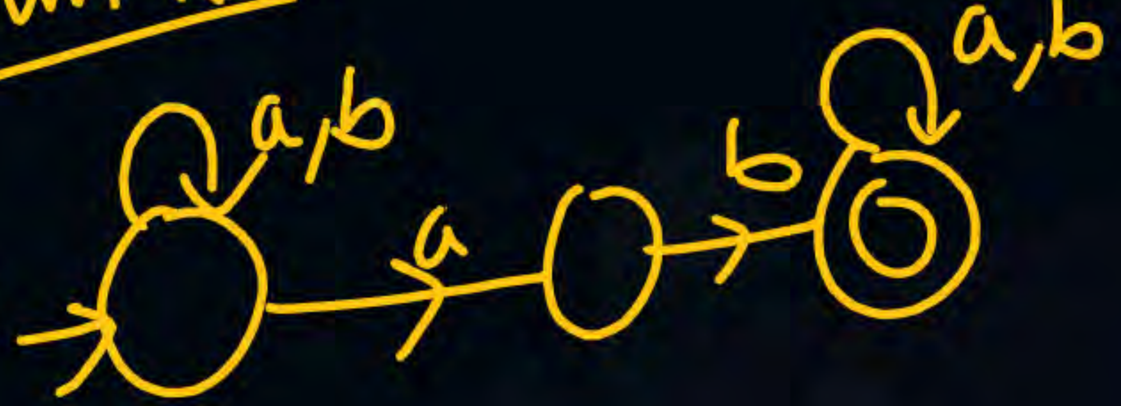


Contains ab as a Substring

Reg Ex :- $(a+b)^* \, a b \, (a+b)^*$

$b^* a a^* b (a+b)^*$

min NFA :

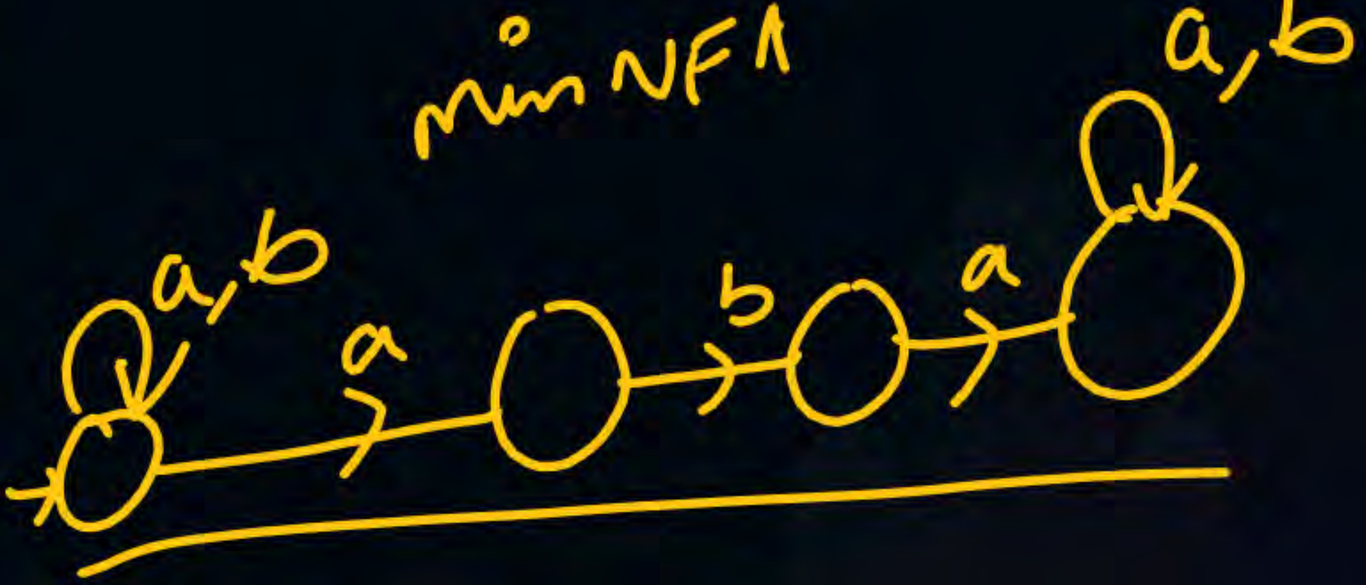


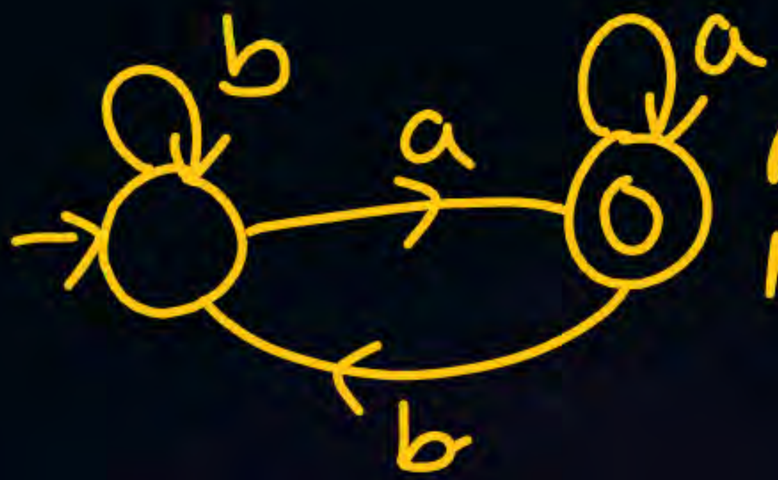
min DFA
min NFA

Contains 'aba' as substring

$$\underbrace{(a+b)^{*}\underline{aba}\,(a+b)^{*}} \rightarrow$$

min NFA

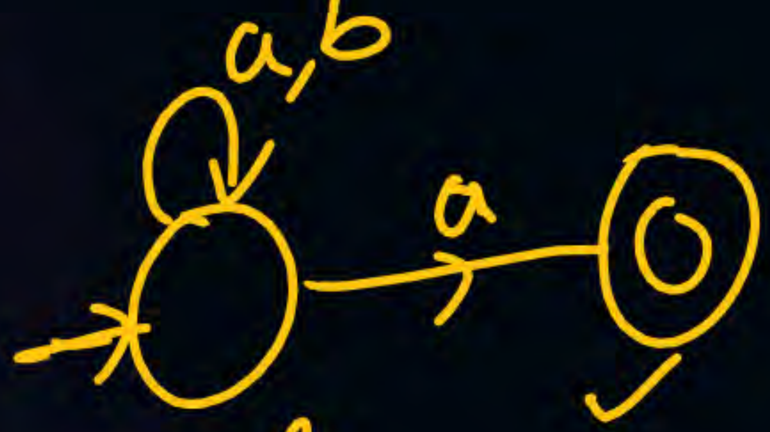




min DFA ✓
min NFA ✓

ends with 'a'

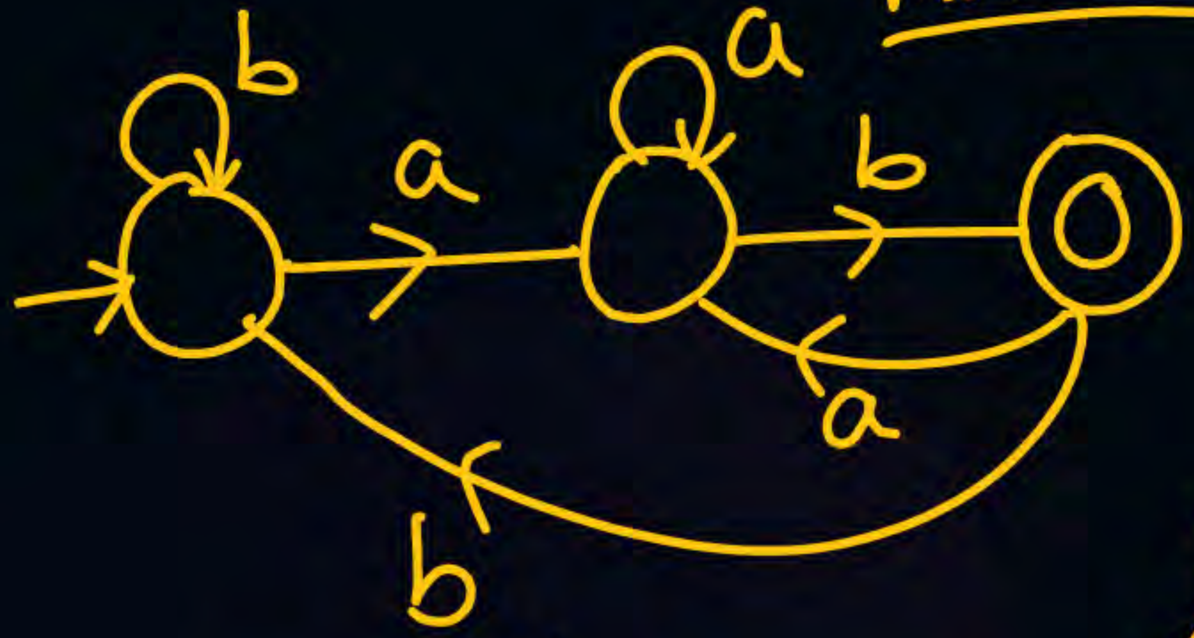
$(a+b)^* a \rightarrow$



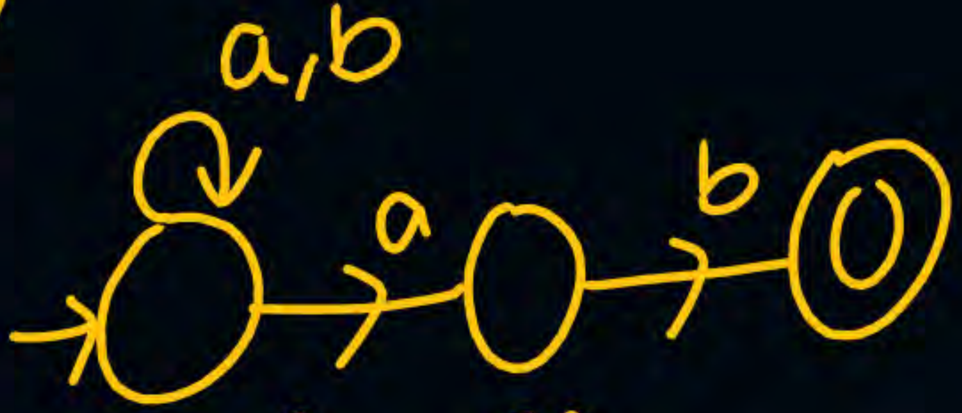
min NFA

min DFA ✓

min NFA ✓

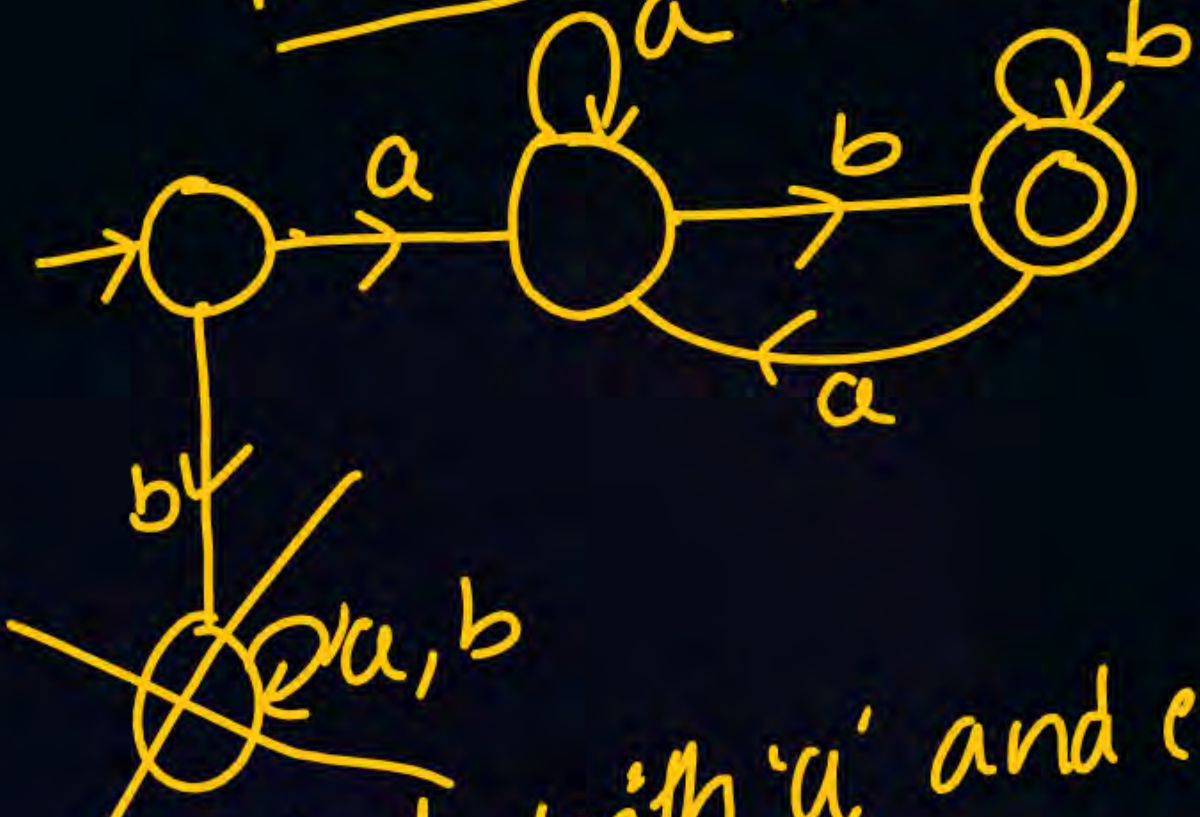


$(a+b)^*ab \longrightarrow$



min NFA

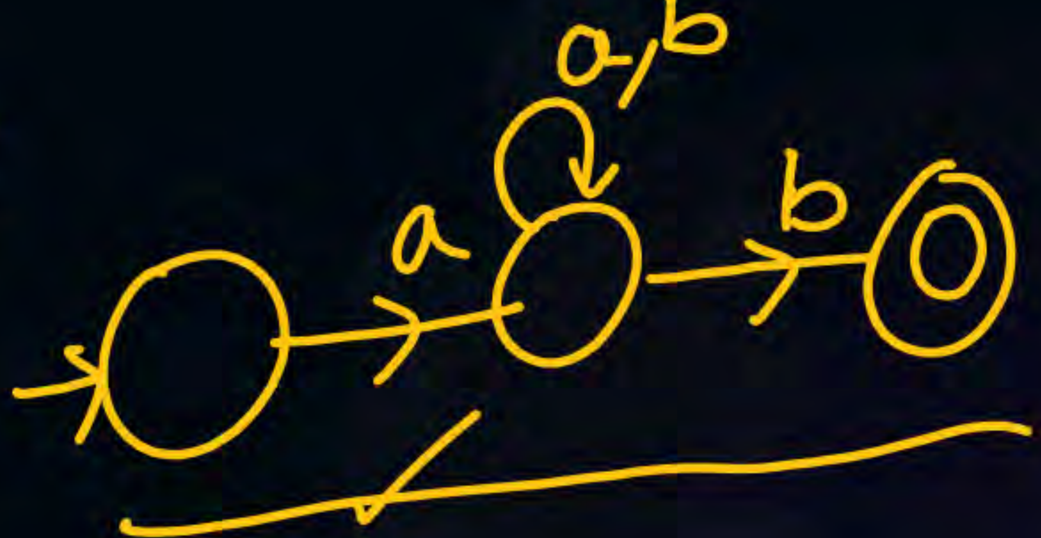
min DFA min NFA



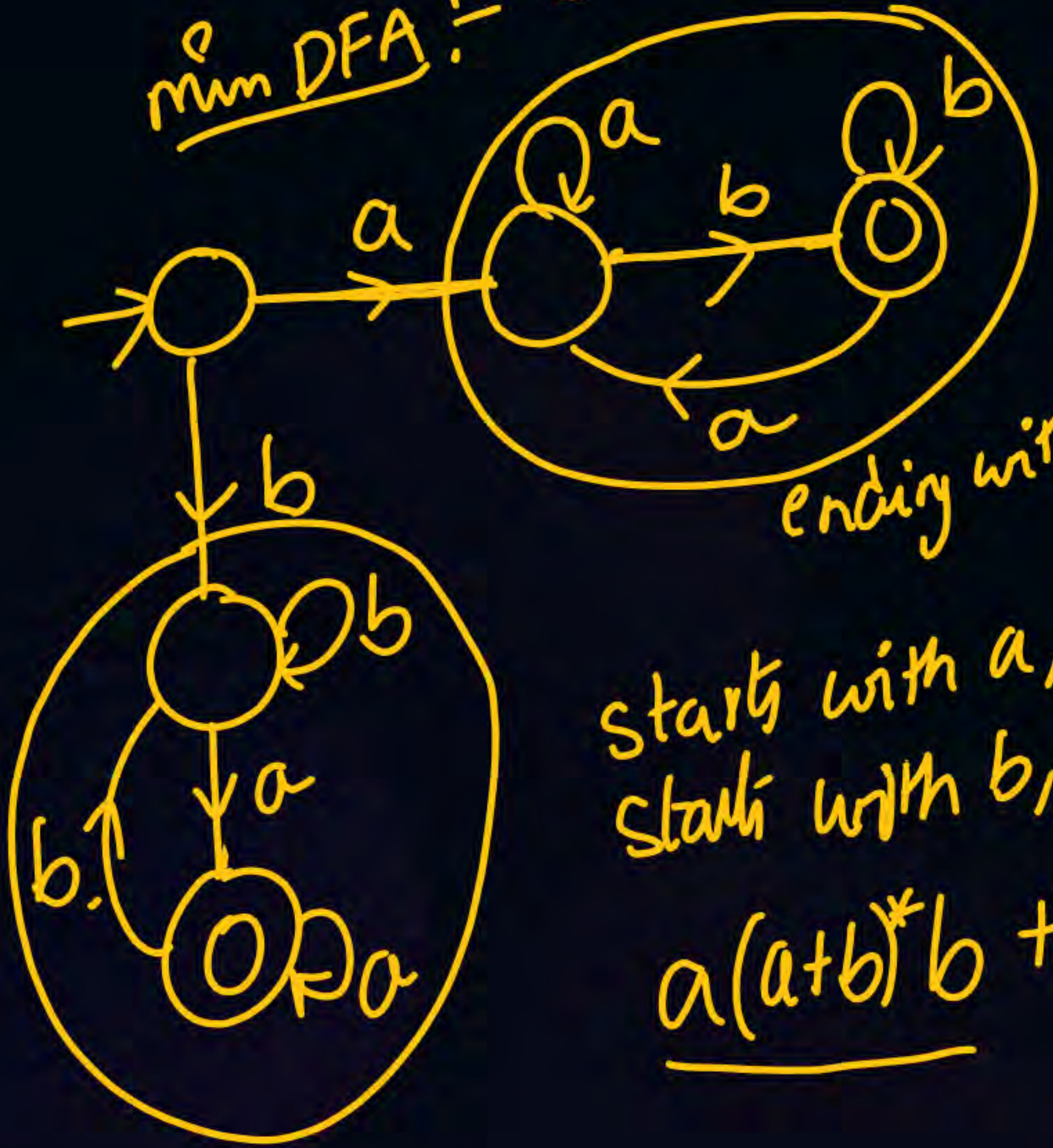
5 minutes

Start with 'a' and end with 'b'

$a(a+b)^*b$ →



min DFA :- (Start and end with diff symbols)



✓ min NFA

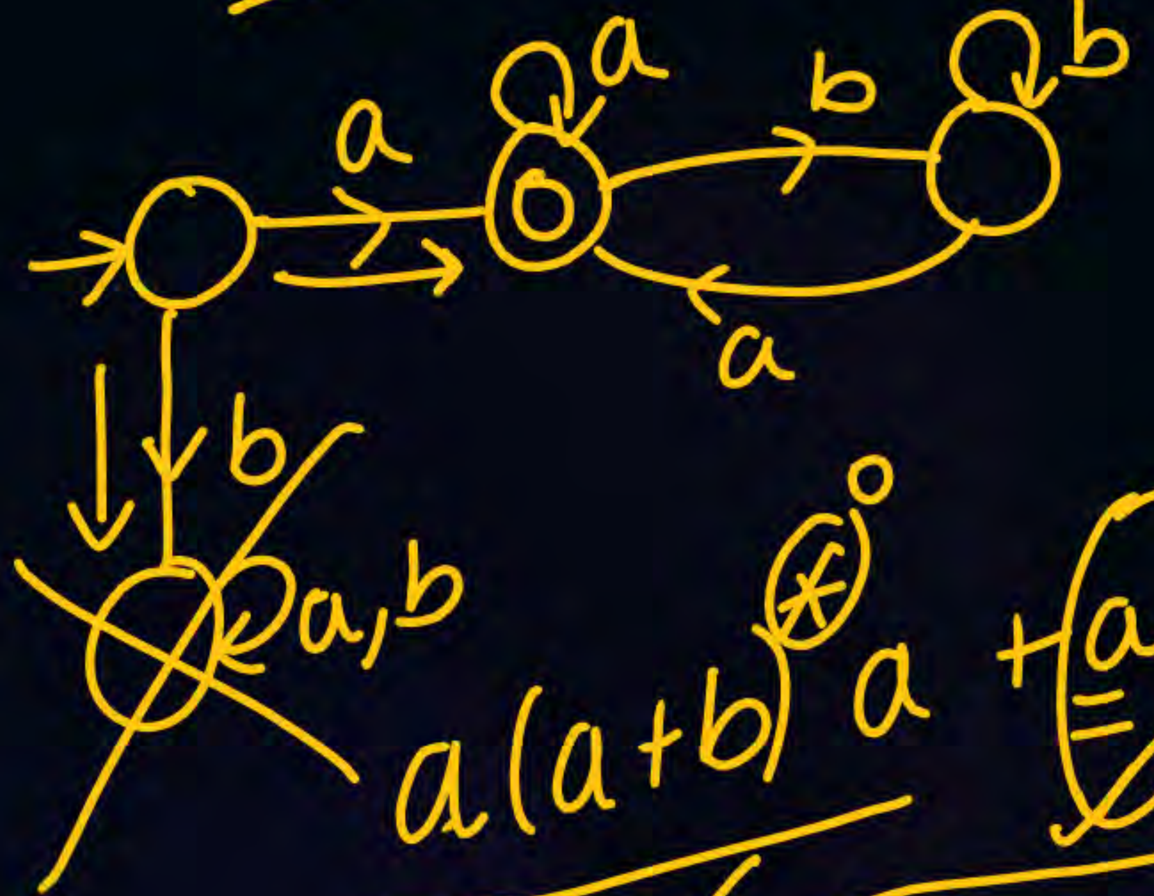
starts with a, ends with b ∪
starts with b, ends with a

$\underline{a(a+b)^*b} + \underline{b(a+b)^*a}$

$(a+b)(a+b)^*(a+b)$

string length at least 2

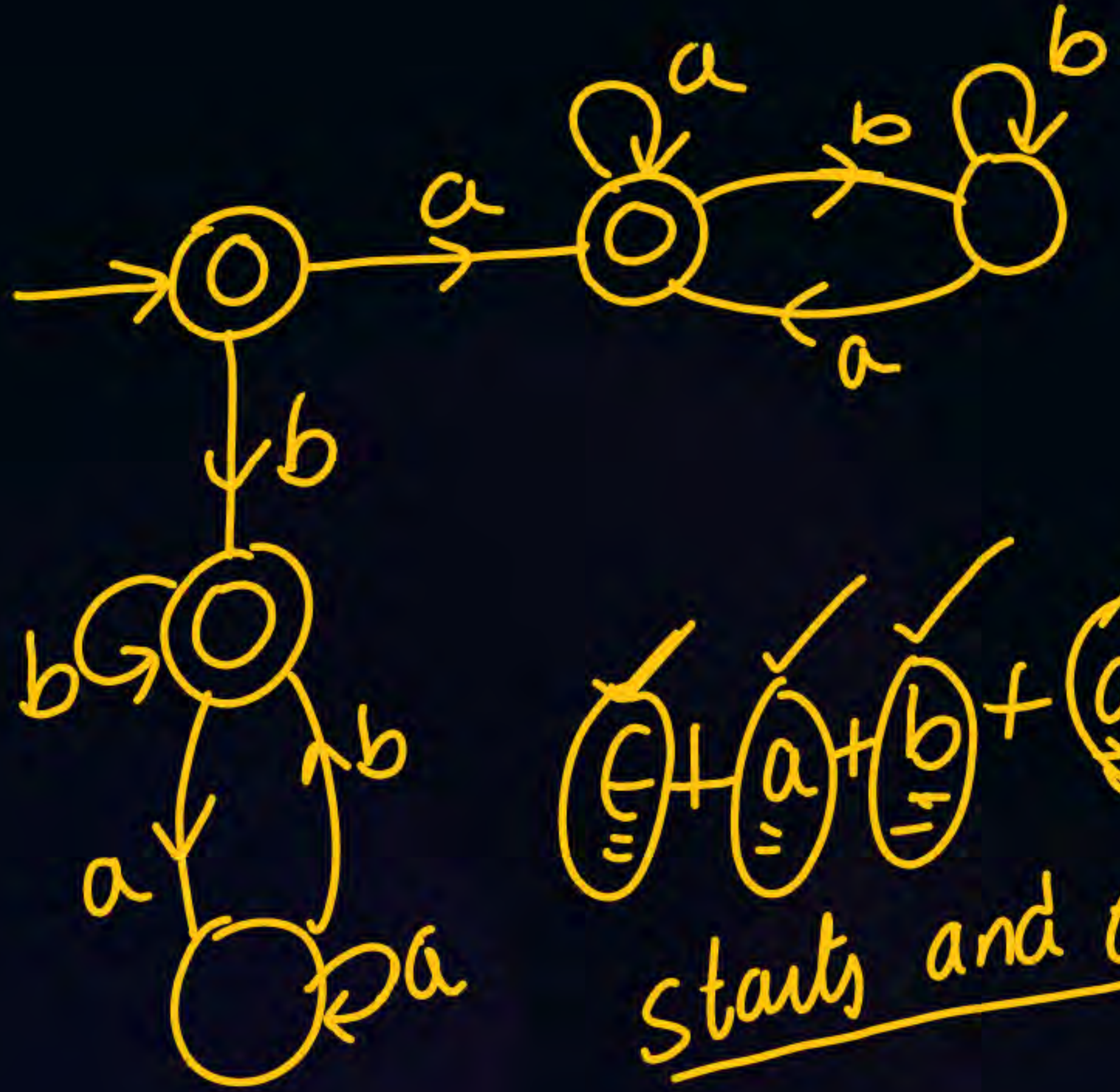
min DFA   min NFA



$a(a+b)^* a + a$

Starting and ending with 'a'

'a' → starts with a and ends 'a'



$$\epsilon + a + b + a(a+b)^*a + b(a+b)^*b$$

starts and ends with same symbol

$$L \longrightarrow DFA \xrightarrow{Com} DFA \longrightarrow \bar{L}$$

$L \downarrow$ Reg $\qquad \bar{L} \downarrow$ Reg

$$\underset{(L)}{\text{min DFA}} \xrightarrow{Comp} \underset{(\bar{L})}{\text{min DFA}}$$

ends with a ✓

$(a+b)^*a$

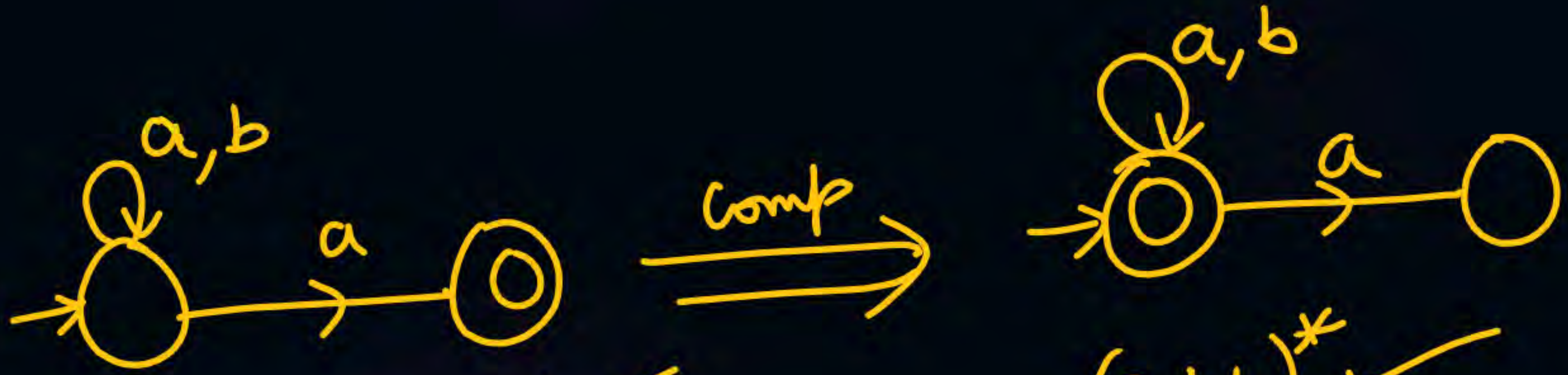
$$\xrightarrow{comp}$$

✓ Not ending with 'a'

$\epsilon + (a+b)^*b$ ✓

FS $\longleftrightarrow$ NFA

Reg languages are closed under complementation



ending with 'a' ✓

$(a+b)^*$ ✓

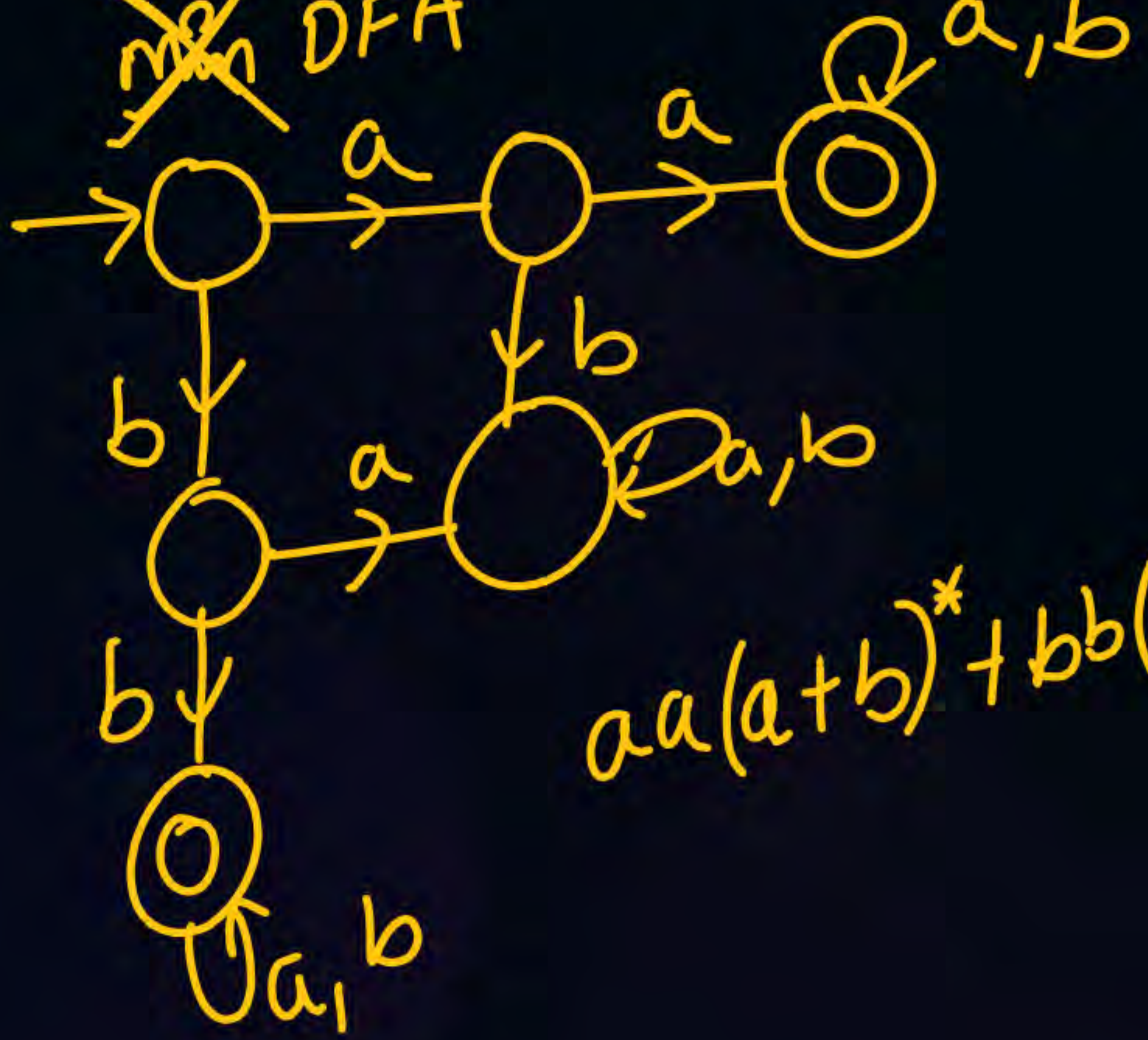
Complementing an NFA, doesn't necessarily complement the language

→ may or may not

if L is reg → $\bar{L}$ is regular

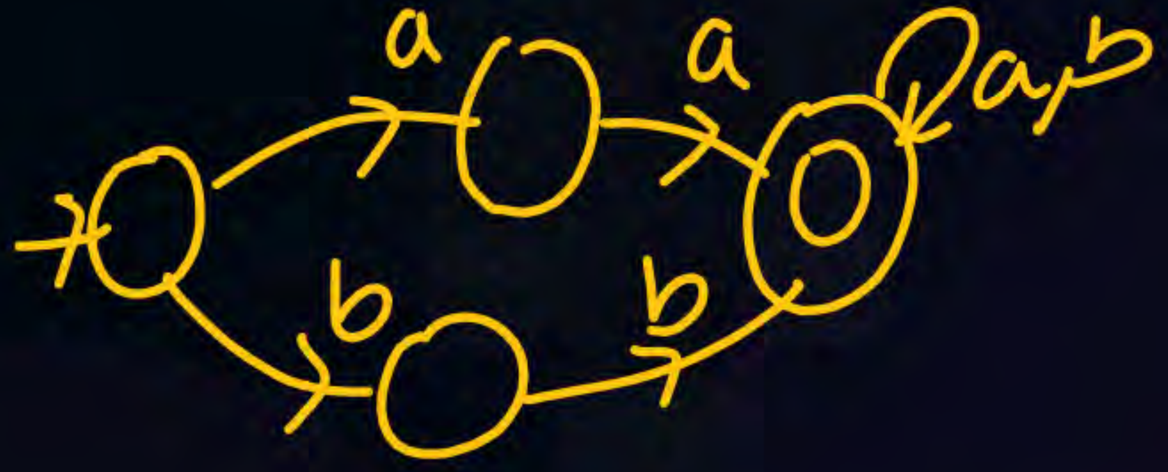
Regular languages are closed under complementation

~~min~~ DFA

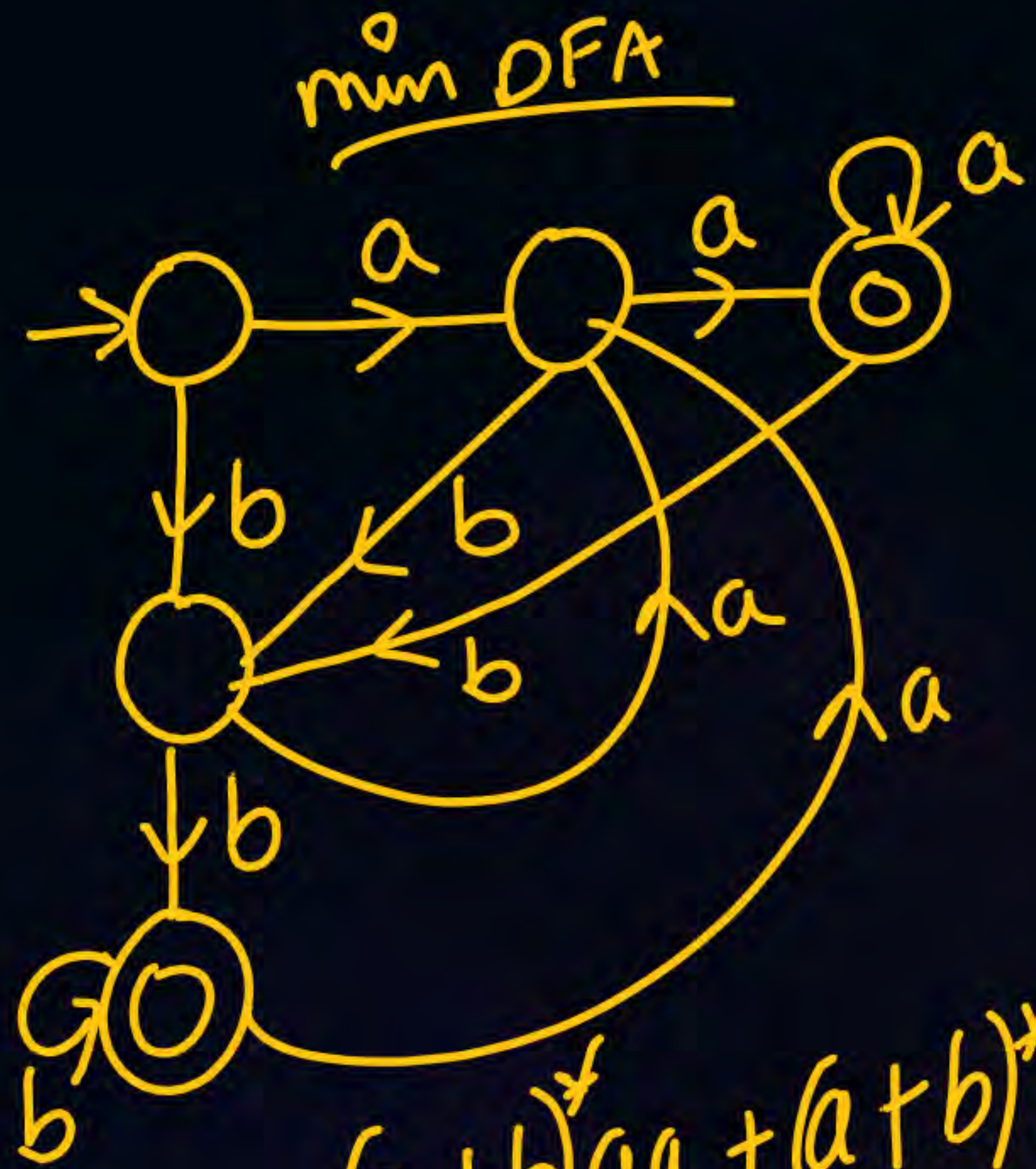


$aa(a+b)^* + bb(a+b)^*$

min DFA

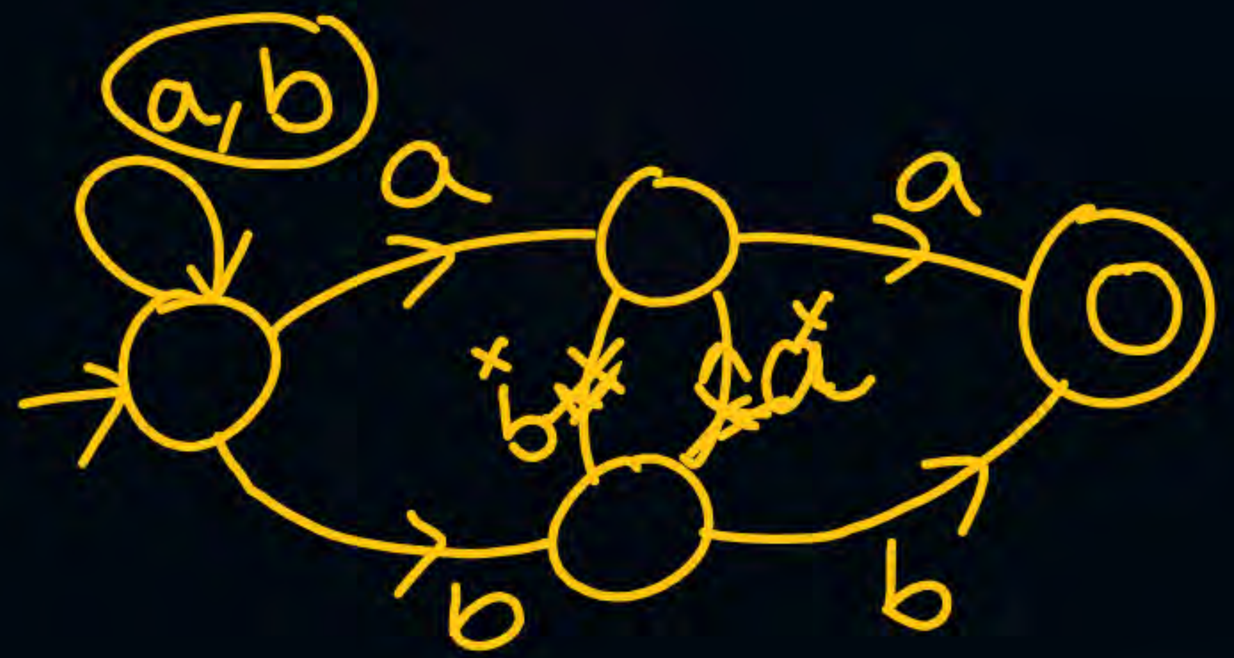


min DFA



$(a+b)^* aa + (a+b)^* bb$

$(a+b)^* (aa+bb)$

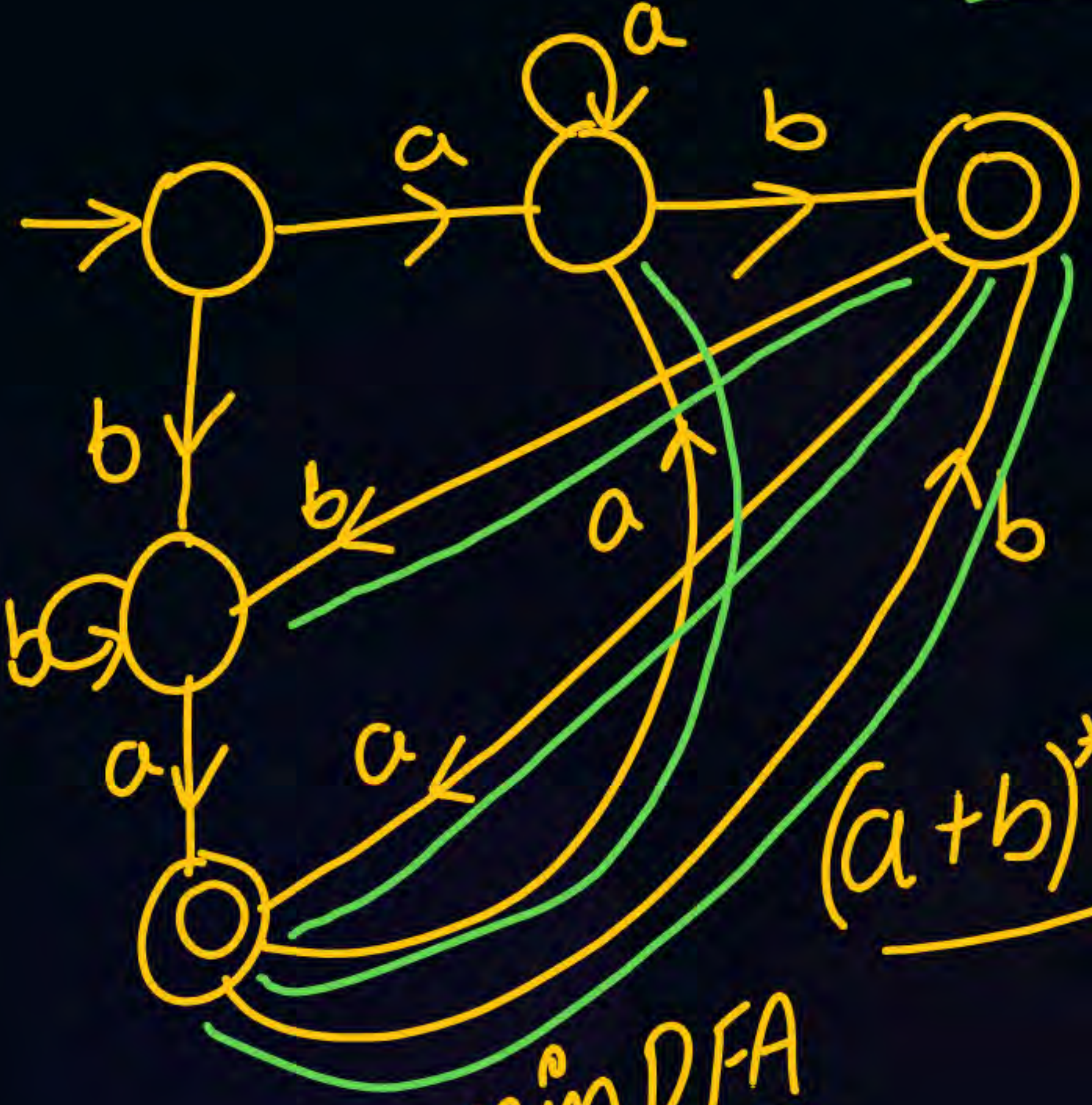


Ending with ba or ab

last two symbols are diff ✓

aba

bab



min DFA

$(a+b)^*(ab+ba)$

min NFA → 4 States

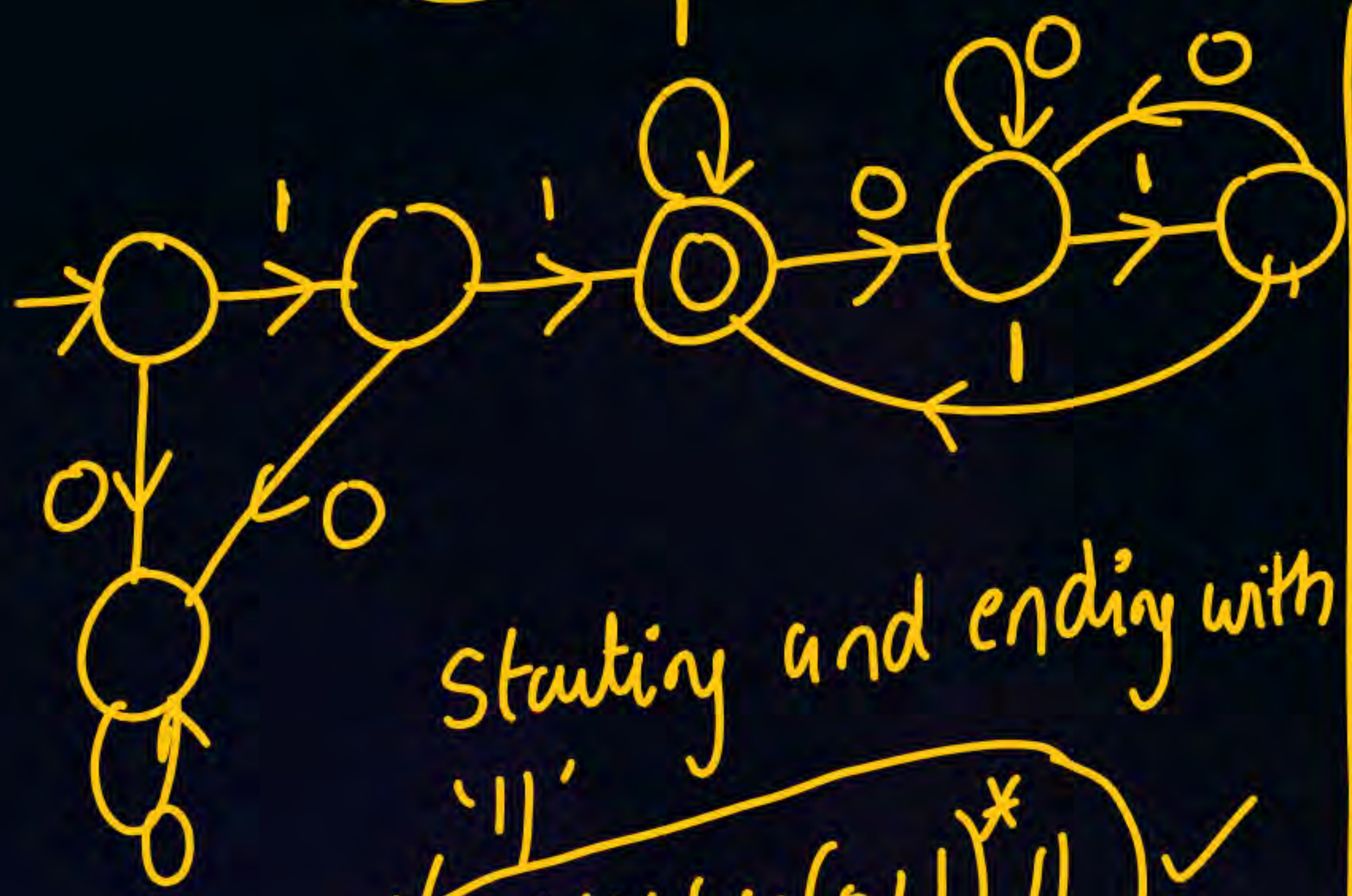
4

min DFA :-

$(ab+ba)(a+b)^*$

min NFA → 4 states

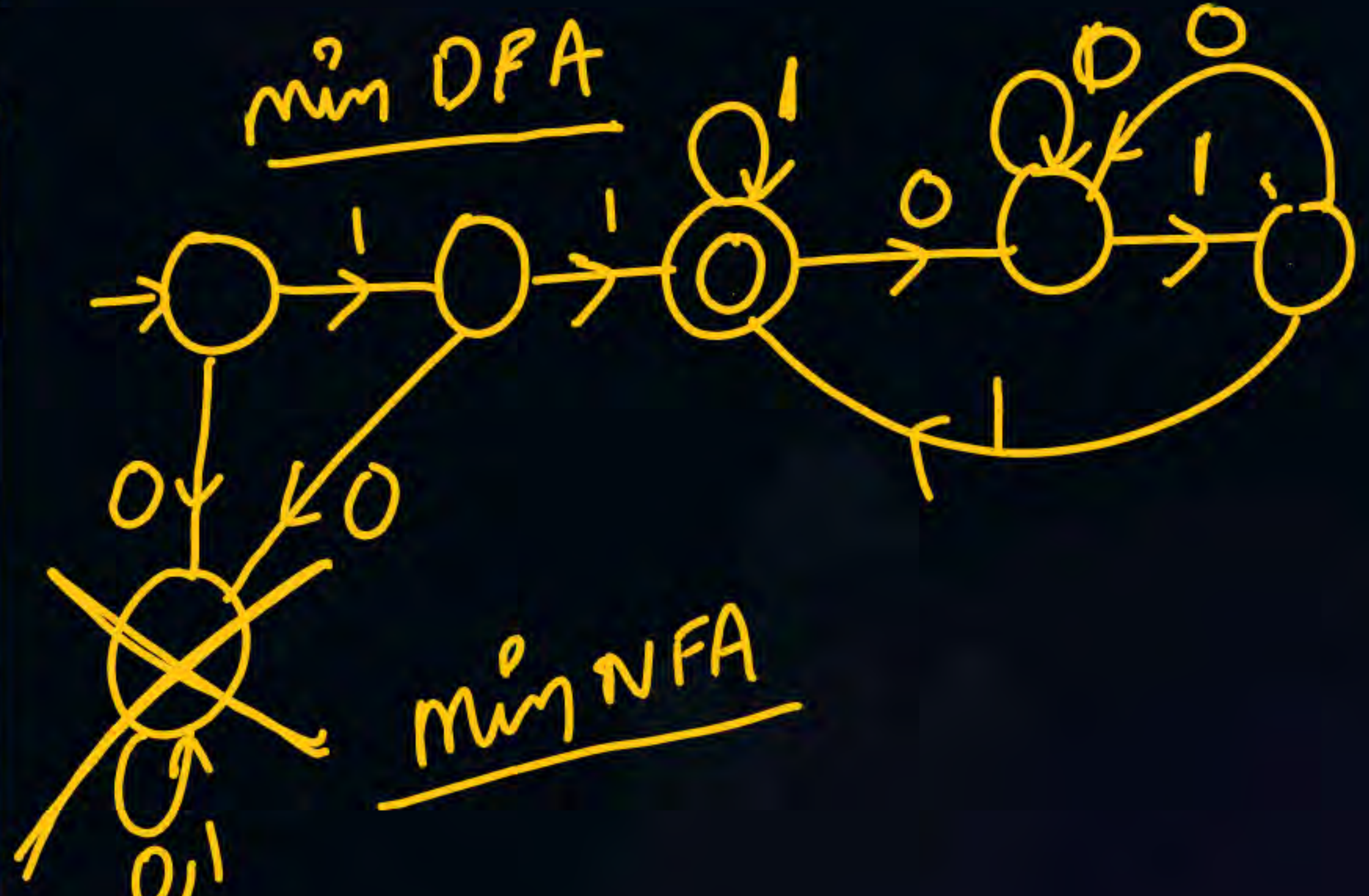
$11 + 111 + 11(0+1)^*11$



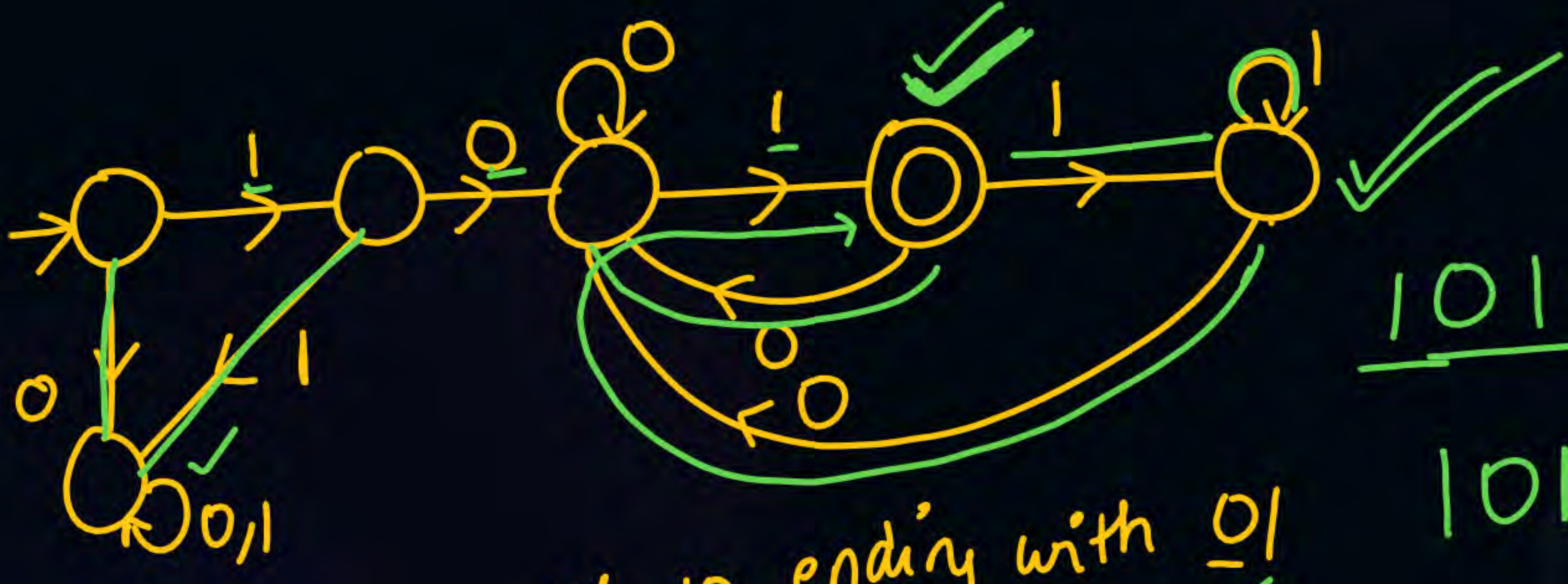
Starting and ending with '11'

$11 + 111 + 11(0+1)^*11$ ✓

min DFA



min NFA



10101

1011101

starting with 10 ending with 01

$101 + 10(0+1)^*01$

1001

THANK - YOU